BRMC Consumer EVT v1.0

Controlled passive-backplane interconnect drawing

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Drawing BRMC-ELEC-EVT-021 | Revision C | Units: mm where dimensional | Date: 2026-09-03

This drawing is the schematic-equivalent connectivity record for the board-only passive backplane. It is generated from the same connector table as the KiCad PCB and is reconciled to the exported IPC-D-356 netlist. KiCad ERC is not applicable because there is no component-level schematic in this artifact.

System interface boundary

OFF-BOARD SOURCE	INTERFACE	BACKPLANE	ENDPOINT MODULES
Mean Well 24 V / 2.5 A	PSM-01 protection + 5/12 V conversion	J_PWR; passive copper; L2/L5 GND	MCU, sensors, buses, AO, safety
Official 27 W USB-C PSU	CM5IO rev2 J11	J_CM5 signal only	CM5 GPIO at CM5IO J8
Display 5 V branch	HARN-04 + DSI FFC	No display power through J_CM5	Waveshare 10.1inch DSI LCD (C)

Power ownership: J_PWR is the only source for 24V_IN, 5V_SYS and 12V_SYS. J_MCU is the sole source for the limited 3V3_SYS interface. CM5IO power remains isolated from all backplane rails.

Connector pin definition

J_CM5 — CM5IO J8 signal-only — Molex 90130-1216

Pin	Net	Function
1	GND	MOLEX_90130-1216_CM5IO_SIGNAL
2	GND	MOLEX_90130-1216_CM5IO_SIGNAL
3	I2C_SCL	MOLEX_90130-1216_CM5IO_SIGNAL
4	I2C_SDA	MOLEX_90130-1216_CM5IO_SIGNAL
5	CM5_TX	MOLEX_90130-1216_CM5IO_SIGNAL
6	CM5_RX	MOLEX_90130-1216_CM5IO_SIGNAL
7	CM5_HEARTBEAT	MOLEX_90130-1216_CM5IO_SIGNAL
8	SAFETY_ACK	MOLEX_90130-1216_CM5IO_SIGNAL
9	SPI_SCK	MOLEX_90130-1216_CM5IO_SIGNAL
10	SPI_MISO	MOLEX_90130-1216_CM5IO_SIGNAL
11	SPI_MOSI	MOLEX_90130-1216_CM5IO_SIGNAL
12	SPI_CS0	MOLEX_90130-1216_CM5IO_SIGNAL
13	GPIO_AUX0	MOLEX_90130-1216_CM5IO_SIGNAL
14	GPIO_AUX1	MOLEX_90130-1216_CM5IO_SIGNAL
15	GND	MOLEX_90130-1216_CM5IO_SIGNAL
16	GND	MOLEX_90130-1216_CM5IO_SIGNAL

J_MCU — safety MCU daughtercard — Molex 90130-1224

Pin	Net	Function
1	5V_SYS	MOLEX_90130-1224_STM32G0B1_CORE
2	GND	MOLEX_90130-1224_STM32G0B1_CORE
3	3V3_SYS	MOLEX_90130-1224_STM32G0B1_CORE
4	CM5_TX	MOLEX_90130-1224_STM32G0B1_CORE
5	CM5_RX	MOLEX_90130-1224_STM32G0B1_CORE
6	CM5_HEARTBEAT	MOLEX_90130-1224_STM32G0B1_CORE
7	SAFETY_ACK	MOLEX_90130-1224_STM32G0B1_CORE
8	I2C_SCL	MOLEX_90130-1224_STM32G0B1_CORE
9	I2C_SDA	MOLEX_90130-1224_STM32G0B1_CORE
10	SPI_SCK	MOLEX_90130-1224_STM32G0B1_CORE
11	SPI_MISO	MOLEX_90130-1224_STM32G0B1_CORE
12	SPI_MOSI	MOLEX_90130-1224_STM32G0B1_CORE
13	SPI_CS0	MOLEX_90130-1224_STM32G0B1_CORE
14	CAN_TX	MOLEX_90130-1224_STM32G0B1_CORE
15	CAN_RX	MOLEX_90130-1224_STM32G0B1_CORE
16	RS485_TX	MOLEX_90130-1224_STM32G0B1_CORE
17	RS485_RX	MOLEX_90130-1224_STM32G0B1_CORE
18	RS485_DE	MOLEX_90130-1224_STM32G0B1_CORE
19	SAFETY_ENABLE	MOLEX_90130-1224_STM32G0B1_CORE
20	TEMP_DATA	MOLEX_90130-1224_STM32G0B1_CORE
21	SWDIO	MOLEX_90130-1224_STM32G0B1_CORE
22	SWCLK	MOLEX_90130-1224_STM32G0B1_CORE
23	NRST	MOLEX_90130-1224_STM32G0B1_CORE
24	GND	MOLEX_90130-1224_STM32G0B1_CORE

Connector pin definition

J_PH — Atlas EZO pH isolated carrier — Molex 22-23-2051

Pin	Net	Function
1	5V_SYS	MOLEX_22-23-2051_ATLAS_EZO_PH_ISO
2	GND	MOLEX_22-23-2051_ATLAS_EZO_PH_ISO
3	I2C_SCL	MOLEX_22-23-2051_ATLAS_EZO_PH_ISO
4	I2C_SDA	MOLEX_22-23-2051_ATLAS_EZO_PH_ISO
5	PH_OFF	MOLEX_22-23-2051_ATLAS_EZO_PH_ISO

J_ORP — Atlas EZO ORP isolated carrier — Molex 22-23-2051

Pin	Net	Function
1	5V_SYS	MOLEX_22-23-2051_ATLAS_EZO_ORP_ISO
2	GND	MOLEX_22-23-2051_ATLAS_EZO_ORP_ISO
3	I2C_SCL	MOLEX_22-23-2051_ATLAS_EZO_ORP_ISO
4	I2C_SDA	MOLEX_22-23-2051_ATLAS_EZO_ORP_ISO
5	ORP_OFF	MOLEX_22-23-2051_ATLAS_EZO_ORP_ISO

J_EC — Atlas EZO EC isolated carrier — Molex 22-23-2051

Pin	Net	Function
1	5V_SYS	MOLEX_22-23-2051_ATLAS_EZO_EC_ISO
2	GND	MOLEX_22-23-2051_ATLAS_EZO_EC_ISO
3	I2C_SCL	MOLEX_22-23-2051_ATLAS_EZO_EC_ISO
4	I2C_SDA	MOLEX_22-23-2051_ATLAS_EZO_EC_ISO
5	EC_OFF	MOLEX_22-23-2051_ATLAS_EZO_EC_ISO

J_TEMP — digital temperature endpoint — Molex 22-23-2041

Pin	Net	Function
1	3V3_SYS	MOLEX_22-23-2041_DIGITAL_TEMP
2	GND	MOLEX_22-23-2041_DIGITAL_TEMP
3	TEMP_DATA	MOLEX_22-23-2041_DIGITAL_TEMP
4	TEMP_AUX	MOLEX_22-23-2041_DIGITAL_TEMP

Connector pin definition

J_PWR — protected 24/5/12 V source — Molex 43045-0600

Pin	Net	Function
1	24V_IN	MOLEX_43045-0600_POWER_HARNESS
2	GND	MOLEX_43045-0600_POWER_HARNESS
3	5V_SYS	MOLEX_43045-0600_POWER_HARNESS
4	GND	MOLEX_43045-0600_POWER_HARNESS
5	12V_SYS	MOLEX_43045-0600_POWER_HARNESS
6	GND	MOLEX_43045-0600_POWER_HARNESS

J_CAN — isolated CAN FD daughtercard — Molex 22-23-2061

Pin	Net	Function
1	5V_SYS	MOLEX_22-23-2061_ISO_CAN_FD_MODULE
2	GND	MOLEX_22-23-2061_ISO_CAN_FD_MODULE
3	CAN_TX	MOLEX_22-23-2061_ISO_CAN_FD_MODULE
4	CAN_RX	MOLEX_22-23-2061_ISO_CAN_FD_MODULE
5	CAN_H	MOLEX_22-23-2061_ISO_CAN_FD_MODULE
6	CAN_L	MOLEX_22-23-2061_ISO_CAN_FD_MODULE

J_485 — isolated RS-485 daughtercard — Molex 22-23-2071

Pin	Net	Function
1	5V_SYS	MOLEX_22-23-2071_ISO_RS485_MODULE
2	GND	MOLEX_22-23-2071_ISO_RS485_MODULE
3	RS485_TX	MOLEX_22-23-2071_ISO_RS485_MODULE
4	RS485_RX	MOLEX_22-23-2071_ISO_RS485_MODULE
5	RS485_DE	MOLEX_22-23-2071_ISO_RS485_MODULE
6	RS485_A	MOLEX_22-23-2071_ISO_RS485_MODULE
7	RS485_B	MOLEX_22-23-2071_ISO_RS485_MODULE

Connector pin definition

J_AO — 8-channel 0-10 V module — Molex 43045-0400

Pin	Net	Function
1	24V_IN	MOLEX_43045-0400_MODBUS_0_10V_8CH
2	GND	MOLEX_43045-0400_MODBUS_0_10V_8CH
3	RS485_A	MOLEX_43045-0400_MODBUS_0_10V_8CH
4	RS485_B	MOLEX_43045-0400_MODBUS_0_10V_8CH

J_PWRMOD — power-module field bus — Molex 43045-0800

Pin	Net	Function
1	24V_IN	MOLEX_43045-0800_POWER_MODULE_BUS
2	GND	MOLEX_43045-0800_POWER_MODULE_BUS
3	CAN_H	MOLEX_43045-0800_POWER_MODULE_BUS
4	CAN_L	MOLEX_43045-0800_POWER_MODULE_BUS
5	RS485_A	MOLEX_43045-0800_POWER_MODULE_BUS
6	RS485_B	MOLEX_43045-0800_POWER_MODULE_BUS
7	SAFETY_ENABLE	MOLEX_43045-0800_POWER_MODULE_BUS
8	GND	MOLEX_43045-0800_POWER_MODULE_BUS

J_SAFE — 24 V safety-relay driver — Molex 22-23-2041

Pin	Net	Function
1	SAFETY_ENABLE	MOLEX_22-23-2041_SAFETY_RELAY_DRIVE
2	GND	MOLEX_22-23-2041_SAFETY_RELAY_DRIVE
3	24V_IN	MOLEX_22-23-2041_SAFETY_RELAY_DRIVE
4	GND	MOLEX_22-23-2041_SAFETY_RELAY_DRIVE

J_SVC — fixture-only SWD/UART — Molex 22-23-2081

Pin	Net	Function
1	3V3_SYS	MOLEX_22-23-2081_SERVICE_DEBUG
2	GND	MOLEX_22-23-2081_SERVICE_DEBUG
3	SWDIO	MOLEX_22-23-2081_SERVICE_DEBUG
4	SWCLK	MOLEX_22-23-2081_SERVICE_DEBUG
5	NRST	MOLEX_22-23-2081_SERVICE_DEBUG
6	CM5_TX	MOLEX_22-23-2081_SERVICE_DEBUG
7	CM5_RX	MOLEX_22-23-2081_SERVICE_DEBUG
8	GND	MOLEX_22-23-2081_SERVICE_DEBUG

Net reconciliation

Net	Domain	Count	Endpoints
12V_SYS	SELV power	1	J_PWR.5
24V_IN	SELV power	4	J_PWR.1 | J_AO.1 | J_PWRMOD.1 | J_SAFE.3
3V3_SYS	logic power	3	J_MCU.3 | J_TEMP.1 | J_SVC.1
5V_SYS	SELV power	7	J_MCU.1 | J_PH.1 | J_ORP.1 | J_EC.1 | J_PWR.3 | J_CAN.1 | J_485.1
CAN_H	field differential	2	J_CAN.5 | J_PWRMOD.3
CAN_L	field differential	2	J_CAN.6 | J_PWRMOD.4
CAN_RX	3.3 V logic	2	J_MCU.15 | J_CAN.4
CAN_TX	3.3 V logic	2	J_MCU.14 | J_CAN.3
CM5_HEARTBEAT	3.3 V logic	2	J_CM5.7 | J_MCU.6
CM5_RX	3.3 V logic	3	J_CM5.6 | J_MCU.5 | J_SVC.7
CM5_TX	3.3 V logic	3	J_CM5.5 | J_MCU.4 | J_SVC.6
EC_OFF	3.3 V logic	1	J_EC.5
GND	common reference	22	J_CM5.1 | J_CM5.2 | J_CM5.15 | J_CM5.16 | J_MCU.2 | J_MCU.24 | J_PH.2 | J_ORP.2 | J_EC.2 | J_TEMP.2 | J_PWR.2 | J_PWR.4 | J_PWR.6 | J_CAN.2 | J_485.2 | J_AO.2 | J_PWRMOD.2 | J_PWRMOD.8 | J_SAFE.2 | J_SAFE.4 | J_SVC.2 | J_SVC.8
GPIO_AUX0	3.3 V logic	1	J_CM5.13
GPIO_AUX1	3.3 V logic	1	J_CM5.14
I2C_SCL	3.3 V logic	5	J_CM5.3 | J_MCU.8 | J_PH.3 | J_ORP.3 | J_EC.3
I2C_SDA	3.3 V logic	5	J_CM5.4 | J_MCU.9 | J_PH.4 | J_ORP.4 | J_EC.4
NRST	3.3 V logic	2	J_MCU.23 | J_SVC.5
ORP_OFF	3.3 V logic	1	J_ORP.5
PH_OFF	3.3 V logic	1	J_PH.5
RS485_A	field differential	3	J_485.6 | J_AO.3 | J_PWRMOD.5
RS485_B	field differential	3	J_485.7 | J_AO.4 | J_PWRMOD.6
RS485_DE	3.3 V logic	2	J_MCU.18 | J_485.5
RS485_RX	3.3 V logic	2	J_MCU.17 | J_485.4
RS485_TX	3.3 V logic	2	J_MCU.16 | J_485.3
SAFETY_ACK	3.3 V logic	2	J_CM5.8 | J_MCU.7
SAFETY_ENABLE	3.3 V logic	3	J_MCU.19 | J_PWRMOD.7 | J_SAFE.1
SPI_CS0	3.3 V logic	2	J_CM5.12 | J_MCU.13
SPI_MISO	3.3 V logic	2	J_CM5.10 | J_MCU.11
SPI_MOSI	3.3 V logic	2	J_CM5.11 | J_MCU.12
SPI_SCK	3.3 V logic	2	J_CM5.9 | J_MCU.10
SWCLK	3.3 V logic	2	J_MCU.22 | J_SVC.4
SWDIO	3.3 V logic	2	J_MCU.21 | J_SVC.3
TEMP_AUX	3.3 V logic	1	J_TEMP.4
TEMP_DATA	3.3 V logic	2	J_MCU.20 | J_TEMP.3